MAT1011
Honors Calculus I
Due: 15 October, 18:00
Submission location: Chengdao 5th

Homework 4

1. Compulsory Exercise

Question 1. Use the linear approximation $(1+u)^r = 1 + ru + o(u)$ as $u \to 0$ to approximate the following functions near $x = 0$. In part (b), give both the tangent-line approximation and the first nonconstant term.

(a) $f(x) = \dfrac{1}{3-x}$;
(b) $f(x) = \sqrt{9+x^2}$;
(c) $f(x) = \sqrt[3]{\dfrac{1+x}{1-x}}$.

Question 2. Assume that g is differentiable wherever the following compositions are defined. Find $f'(x)$.

(a) $f(x) = g(x^2 + g(x))$;
(b) $f(x) = g\big(x[g(x)]^2\big)$;
(c) $f(x) = g(g(x^2))$.

Question 3. Find the following limits **using linear approximation of functions**. **Do not use L'Hopital's rule.** For part (b), first set $t = 1/x$; in part (c), apply the approximations to the inner functions as well as to the square roots.

(a) $\displaystyle\lim_{x\to 0} \frac{\sqrt{1+4x} - \sqrt[3]{1+3x}}{x}$;
(b) $\displaystyle\lim_{x\to+\infty} \left(\sqrt[n]{x^n + x^{n-1}} - x\right)$, where $n \geq 1$ is an integer;
(c) $\displaystyle\lim_{x\to 0} \frac{\sqrt{1+\sin x} - \sqrt{1-\sin x}}{\tan(2x)}$.

Question 4. Suppose y is implicitly defined by

$$x^2 + xy + y^2 = 7.$$

Find $y' = dy/dx$ and $y'' = d^2y/dx^2$ at points where the required denominators are nonzero.

Question 5. Prove that

$$|\arctan x - \arctan y| \leq |x - y| \qquad \text{for all } x, y \in \mathbb{R}.$$

Question 6. Let $f(x) = x^4 - 4x^3$ for $x \in \mathbb{R}$. Determine the intervals on which its graph is concave up or concave down, and find the coordinates of all inflection points. Justify each inflection point by checking that the concavity changes.

Question 7. Suppose $f : (1, +\infty) \to \mathbb{R}$ is differentiable and

$$\lim_{x\to+\infty} f'(x) = L \in \mathbb{R}.$$

Prove that $\displaystyle\lim_{x\to+\infty} f(x)/x = L$.

Question 8. Let $f : [0,1] \to \mathbb{R}$ be continuous on $[0,1]$, differentiable on $(0,1)$, and satisfy $f(0) = 0$. Suppose that for some constant $C > 0$,

$$|f'(x)| \leq C|f(x)| \qquad (0 < x < 1).$$

Prove that $f \equiv 0$ on $[0,1]$.

Question 9. Use the Intermediate Value Theorem and the Mean Value Theorem (including Rolle's theorem as a special case) to prove that the equation

$$x^3 - 3x + 1 = 0$$

has exactly three distinct real solutions. Locate one solution in each of $(-2,-1)$, $(0,1)$ and $(1,2)$, and prove that there are no other real solutions. You do not need to find the solutions explicitly.

2. Extra Exercises

Question 10. Prove that, for $x, y > 0$ and rational numbers $\alpha, \beta \in \mathbb{Q}$ with $0 < \alpha < \beta$,

$$(x^\alpha + y^\alpha)^{1/\alpha} > (x^\beta + y^\beta)^{1/\beta}.$$

All powers are taken on positive bases.

Question 11. Let f be differentiable on an open interval containing $[a,b]$, where $a < b$, and suppose

$$f'(a) = f'(b).$$

Prove that there exists $c \in (a,b)$ such that

$$f'(c) = \frac{f(c) - f(a)}{c - a}.$$

3. Thomas Textbook Exercises

Chapter 1. Limits and Differentiation
Section 1.2. Derivatives
§1.2.3. Differentials: Exercises 3.9: 14(f), 38, 56.
Chapter 3 Practice Exercises: 14, 36, 40, 118.
Chapter 3 Additional and Advanced Exercises: 4, 20, 21, 24.
§1.2.4. Extreme values of functions: Exercises 4.1: 56, 62, 68; Exercises 4.5: 28.
§1.2.5. The mean value theorem: Exercises 4.2: 16, 56, 66, 72.